Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., III & CSBS
Course Code & Title: AD3351 Design and Analysis of Algorithms
Name of the Faculty member: Ms.K.Usharani AP/CSBS

4.10.23

Innovative Practice Description

- **Unit / Topic:** Unit 2 / Knapsack Problem
- **Course Outcome:** CO 2 – Analyze the efficiency of brute force, divide and conquer, decrease and conquer, Transform and conquer algorithmic techniques
- **Topic Learning Outcome:** Analyzing the efficiency of brute force technique by string matching, TSP, Knapsack problem and Assignment problem
- **Activity Chosen:** Active Learning: Shower Board
- **Justification:**
    1. Shower Board is an active learning and also a collaborative learning method which increase the students focus and involvement on learning the topic.
    2. Sharing their knowledge with their teams will improve their understanding in the topic and also their communication skill.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
    1. **Material used :** Activity Sheet to write the answers . Students has to find answers for the given question in ppt. And also they have to justify their answers to their friends.
    2. **Prerequisite :** Basic logical thinking ability.
    3. Activity comprises of 4 parts – Understanding the problem, Question View , Showcasing the answer in front of their friends, Finalize the correct answer.
    4. **Understanding the problem** – The faculty explained the constraints of Knapsack problem and its objective functions.
    5. **Question View** – Questions shown to students commonly.
    6. **Showcase-** Students are asked to write their answers in their sheets. Students are combined as group of 2 members to discuss their knowledge and also to get clarity on the topic. They are asked to fill the worksheet after discussing with their team members.
    7. **Finalizing the answers** – After discussion, the correct answer is posted in the board.

- **CO – PO / PSO mapping:**

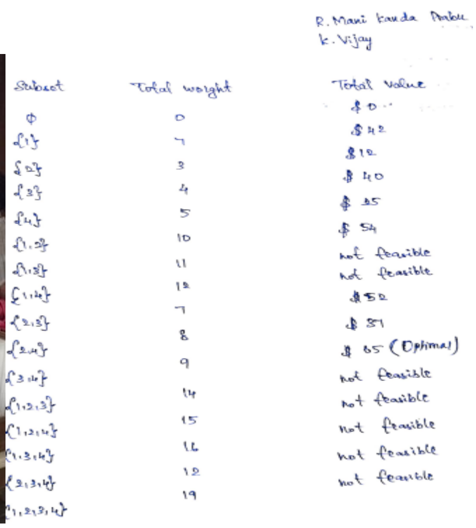
CO	PO1	PO2	PO3	PO9	PO12	PSO3
CO5	2	1	1	3	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO12**	**PSO3**
	2	**1**	**1**	**3**	**2**	**3**
Justification for correlation	Apply the knowledge of engineering fundamentals like concepts of algorithms.	Identify and analyze complex engineering problems related to optimization problems.	Design solutions for complex engineering problems	Function effectively as an individual and as a member in a team	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning.	Apply appreciable knowledge of management skills cultured through real time practice.

- **Images / Screenshot of the practice:**



EXHAUSTIVE SEARCH
KNAPSACK PROBLEM

Gayathri Meena G
Shaly S

Subset	Total Weight	Total Value
{1, 2}	10	$54
{1, 3}	11	$82 (not feasible)
{1, 4}	12	$67 (not feasible)
{2, 3}	7	$52
{2, 4}	8	$37
{3, 4}	9	$65
{1, 2, 3}	14	$94 (not feasible)
{1, 2, 4}	15	$79 (not feasible)
{1, 3, 4}	16	$107 (not feasible)
{2, 3, 4}	12	$77 (not feasible)

∴ {3, 4} ⇒ 65   9 kg.

EXHAUSTIVE SEARCH
KNAPSACK PROBLEM

M· Archana
K·T· Akshaya

{1, 2}	10	54
{1, 3}	11	82
{1, 4}	12	67
{2, 3}	7	52
{2, 4}	8	37
{3, 4}	9	65
{1, 2, 3}	14	94
{1, 2, 4}	15	79
{1, 3, 4}	16	107
{2, 3, 4}	12	77
{1, 2, 3, 4}	19	119

∅	0	0
1	7	42
2	3	12
3	4	40
4	5	25

∴ {3, 4} ⇒ 65   9 kg

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends.
  - ***Benefit of the practice:***

    Students were actively participated in this activity.

    Students interaction with their peer members and the class during the activity was good and effective.
  - ***Challenges faced in implementation:***

    Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

1. https://www.celt.iastate.edu/wp-content/uploads/2017/03/CELT226active learningtechniques.pdf